

CWC National Flood Forecasting Network

Flood Forecasting Services

Flood causes considerable damage to human lives and property almost every year. Since the adoption of National Flood Policy by Government of India in 1954, it was realized that a total protection against flood by structural means alone is not possible and that optimum solution would consist of a mixture of structural and non-structural measures. Therefore, stress has been laid on non-structural measures like flood forecasting and warning, which is most important among such means to minimize the damage potential from floods. Accurate and timely flood forecasts and advance warning have, therefore, to be aimed for providing valuable time to the people and to civil authorities in taking preventive measures like evacuation, relief and rehabilitation measures, preparedness for flood fighting by engineering authorities etc. and thus mitigating such losses from floods.

Flood Forecasting Network in the Country

Flood Forecasting has been recognized as the most important, reliable and cost effective non-structural measures for flood mitigation. Recognizing the great importance of this measure, flood forecasting of river Yamuna at Delhi was suggested by Reddy Committee set up by Prime Minister, Govt. of India to manage flooding of Delhi. Accordingly in the year 1958, CWC commenced the flood forecasting service in a small way by establishing flood forecasting unit for issuing water level forecasts of the Yamuna for the National Capital, Delhi. On the recommendation of various committees/panels, a "Flood Forecast & Warning Organisation" was set up in CWC in 1969 to establish forecasting sites on inter-state rivers at various flood prone places in the country. 41 forecasting sites were added in 1969, making total number of forecasting sites to 43. The Salient Features of Flood Forecasting System is given in the following Table:

1.	Establishment of 'First Scientific Flood Forecasting Unit' (F.F.U.) at Delhi in India	November, 1958
2.	Date of issue of first scientific flood forecast	25 th July, 1959
3.	Name of first forecasting site and river	Delhi Railway Bridge (old) on the River Yamuna
4.	Year of commencement of flood forecasting system on the inter-state rivers i.e. first national level expansion	1969
5.	No. of Chief Engineer's offices including one CE (Flood Management) at CWC' headquarters, Monitoring – Central, Nagpur and Cauvery and Southern Rivers Organisation, Coimbatore being organisations supporting the Flood Forecasting Activities	11
6.	No. of Superintending Engineer's offices including one Flood Forecast Monitoring Directorate at CWC headquarter	14
7.	No. of present Flood Forecasting Divisions and No. of Divisions supporting FF Activities	25
8.	No. of Control Room/Sub-Divisions engaged in flood forecasting work	64
9.	No. of inter-state rivers (main/tributaries) covered by flood forecasting programme	71
10.	No. of states including union -territories covered under F.F. Programme	17
11.	No. of forecasting sites	175
12.	No. of exclusive base stations	165
13.	No. of gauge and gauge & discharge sites	866
14.	No. of exclusive rain gauge stations (ordinary/self recording)	97
15.	No. of wireless stations including Control Rooms	544
16.	No. of Telemetry Stations installed/under installation during IX, X and XI Plans	445
17.	Maximum no. of forecasts issued in any one year Second Highest no. of forecasts issued	8566 (in 1990) 8223 (in 2007)
18.	No. of forecasts issued in flood season 2009	4010
19.	No. of forecasts issued in flood season 2010	7519
20.	No. of forecasts issued in flood season 2011	5991
21.	No. of forecasts issued in flood season 2012	5031

The "National Flood Forecasting and Warning Network" of Central Water Commission, which comprised of 175 flood forecasting sites including 28 inflow forecasting sites in flood season 2012 is shown in Map-1. The number of flood forecasting sites on each of the nine major inter-state river systems, which constitutes 71 river sub-basins in the country, are given in the Table 1.1.

Table 1.1: Number of flood forecasting sites in major inter-state river systems

S. No.	Major Interstate River Systems	Type of Forecasting Sites		Total
		Level Forecasting	Inflow Forecasting	
1	Ganga & its tributaries	77	10	87
2	Brahmaputra & its tributaries	27	00	27
3	Barak System	05	00	05
4	Eastern Rivers	08	01	09
5	Mahanadi	03	01	04
6	Godavari	14	04	18
7	Krishna	03	06	10
8	West Flowing Rivers	09	06	15
9	Southern River System (Pennar)	01	00	01
Total		147	28	175

The above flood forecasting network covers the following 15 states, one Union Territory and NCT of Delhi as shown in the Table 1.2 Table 1.2 State-wise Flood Forecasting Network in CWC

Sl. No.	State	Type of Forecasting sites		Total Forecasting sites
		Stage forecasting	Inflow forecasting	
1	Andhra Pradesh	9	7	16
2	Assam	24	0	24
3	Bihar	32	0	32
4	Chhattisgarh	1	0	1
5	Gujarat	6	5	11
6	Haryana	0	1	1
7	Jharkhand	1	4	5
8	Karnataka	1	3	4
9	Madhya Pradesh	2	1	3
10	Maharashtra	7	2	9
11	Orissa	11	1	12
12	Tripura	2	0	2
13	Uttarakhand	3	0	3
14	Uttar Pradesh	34	1	35
15	West Bengal	11	3	14
16	Dadra & Nagar Haveli	1	0	1
17	NCT of Delhi	2	0	2
Total		147	28	175

A list of 175 FF Stations is at Annex-I.

Central Water Commission through its twenty flood forecasting divisions issued forecasts to the various user agencies, which includes various civil / engineering agencies of the States/ Central Governments such as Irrigation/ Revenue/ Railways/ public undertakings and Dam/ Barrage Authorities/ District Magistrates/ Sub Divisional Officers besides the Defence Authorities involved in the flood loss mitigation work. During the flood season, the Hon'ble Minister of Water Resources, Government of India, the Chairman and the Member (River Management) of Central Water commission were also being apprised of the latest flood situations in the above river basins in the country.

Classifications of Various Flood Situations

The Central Water Commission has categorized various flood situations, for monitoring the floods in the country through its flood forecasting network, into the following four different categories, depending upon the severity of floods i.e. based on floods magnitudes.

Level Forecast

****Low Flood****

The river is said to be in "LOW FLOOD" situation at any flood forecasting sites when the water level of the river touches or crosses the warning level, but remains below the danger level of the forecasting site.

Moderate Flood

If the water level of the river touches or crosses its danger level, but remains 0.50 m below the Highest Flood Level of the site (commonly known as "HFL") then the flood situation is called the "MODERATE FLOOD" situation.

High Flood

If the water level of the river at the forecasting site is below the Highest Flood Level of the forecasting site but still within 0.50m of the HFL then the flood situation is called "HIGH FLOOD" situation. In "High Flood Situations" a special "**Orange Bulletin**" is being issued by the Central Water Commission to the users agencies which contains the "special flood message" related to the high flood.

Unprecedented Flood

The flood situation is said to be "**UNPRECEDENTED**" when the water level of the river crosses the "HIGHEST FLOOD LEVEL" recorded at any forecasting site so far. In "Unprecedented Flood Situations" a special "**Red Bulletin**" is being issued by the Central Water Commission to the users agencies which contains the "special flood message" related to the unprecedented flood.

Inflow Forecasts

Inflow Forecasts are issued for 28 dams/reservoirs/barrages in various river basins in the country. The project authorities have identified the threshold inflow limits for issue of forecast considering various factors such as safety of the dam, status of reservoir, downstream channel/ canal requirements.

Standard Operating Procedure (SOP) for Flood Forecasting & Warning

The basic activity of data collection, its transmission and dissemination of flood forecasts to the local administration is carried out by the field divisions of CWC. The modelling centres and Divisional Flood Control Rooms (DFCR) are located in the premises of the field divisions. The field divisions perform these activities as per existing Manual on Flood Forecasting which contains the following critical activities as the general SOPs

1. Nomination of Nodal Officers of CWC for interaction with the Nodal Officers of concerned State Governments before monsoon every year.
2. Gearing up of flood forecasting network before monsoon every year.
3. Operation of Divisional Flood Control Room during monsoon every year
4. Operation of Central Flood Control Room (CFCR) during monsoon every year.
5. Issue of flood forecasts to designated officers of concerned State and transmission thereof through FAX/Telephone/E-mail/ through Special Messengers during monsoon every year.
6. Sending flood alerts through SMS on Mobile Phones to the concerned officers of State/ Central Government during high and unprecedented flood situations as per Standard Operating procedure (SOP) for issuing alerts and electronic messaging in the event of disaster situations issued by National Disaster Management Division, Ministry of Home Affairs, vide letter No: 31-32/2003-NDM-III / II dated 10th April 2006, made effective from 24th April 2006.

For the purpose of dissemination of alerts to PMO/ Cabinet Secretariat, a uniform system has been devised by categorizing each type of alert in stages- Yellow, Orange and Red. Categories of alerts for flood in respect of level forecasts is as indicated below.

Sl. No.	State	Type of Forecasting sites		Total Forecasting sites
		Stage forecasting	Inflow forecasting	
1	Andhra Pradesh	9	7	16
2	Assam	24	0	24
3	Bihar	32	0	32
4	Chhattisgarh	1	0	1
5	Gujarat	6	5	11
6	Haryana	0	1	1
7	Jharkhand	1	4	5
8	Karnataka	1	3	4
9	Madhya Pradesh	2	1	3
10	Maharashtra	7	2	9
11	Orissa	11	1	12
12	Tripura	2	0	2
13	Uttarakhand	3	0	3
14	Uttar Pradesh	34	1	35
15	West Bengal	11	3	14
16	Dadra & Nagar Haveli	1	0	1
17	NCT of Delhi	2	0	2
Total		147	28	175

Modernisation

In order to meet the requirement of real-time data collection, automatic data transmission and flood forecast formulation, expeditious data / information dissemination, the Central Water Commission has undertaken modernization of its data collection and flood forecast network. During IX Plan, 55 telemetry stations in Mahanadi and Chambal Basins and two Earth Receiving Stations (ERS) at Jaipur and Burla were established. During X Plan, modernization of 168 stations and 11 Modeling Centers was undertaken which has been completed in XI Plan. During XI Plan, additional 222 telemetry stations, 10 Modeling Centers and one more ERS at New Delhi have been installed. During 12th Plan, remaining stations are proposed to be modernized. New stations proposed during 12th Plan, are proposed to be equipped with modern equipment.

Expansion of FF Network

Extension of the service followed from time to time and now the river forecasting has been expanded over the years to cover nine major inter-state flood prone river basins, which comprises of 71 sub-river basins traversing the country. The year-wise positions of the number of flood forecasting sites till the flood season 2012 in the network of Central Water Commission are shown in the Table 1.3:

Year	No. of Flood Forecasting Sites	Year	No. of Flood Forecasting Sites
1958	01	2001	159
1965	02	2002	161
1969	43	2003	166
1977	77	2004	172
1980	84	2005	173
1985	145	2006	175
1987	147	2007	175
1990	157	2011	175

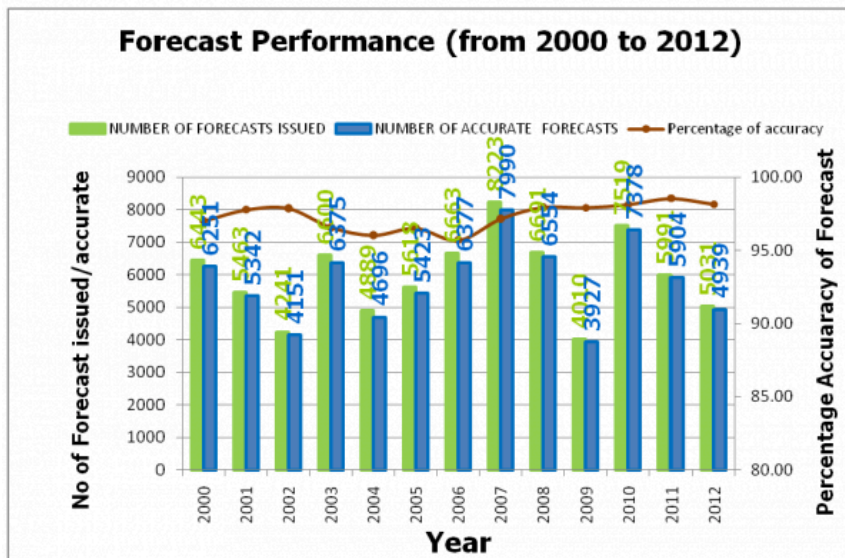
Table-1.3: Yearwise positions of number of forecasting sites in CWC

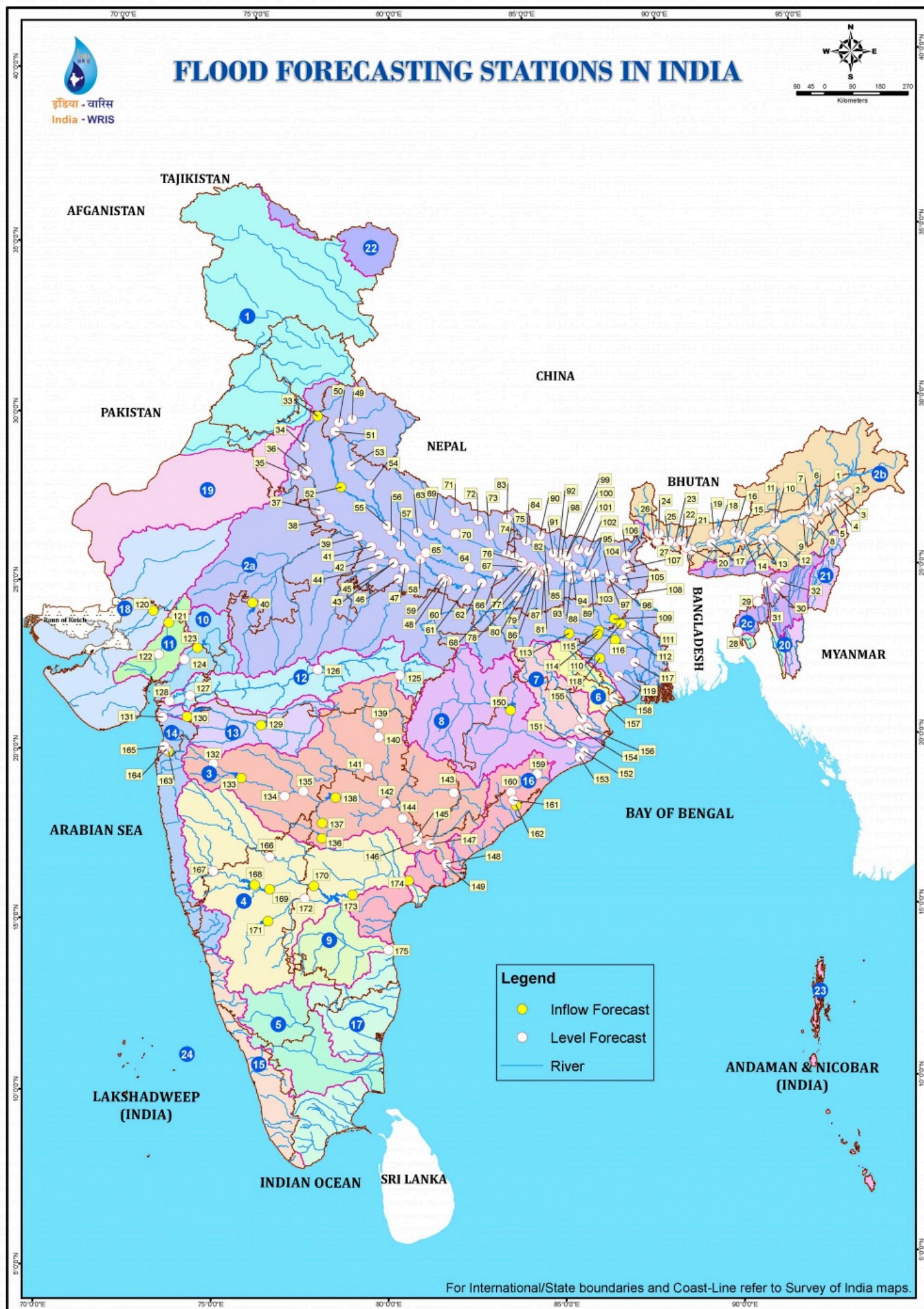
Performance of Flood Forecasting Network

A number of techniques are being utilised for formulation of river stage and inflow forecasts by Central Water Commission. While inflow forecast is being provided for assisting project authorities in reservoir regulation, the stage forecast is done for warning the civil and engineering authorities about the predicted water level well ahead of its occurrence. An accurate forecast is one where the forecast level and corresponding actual observed level exactly synchronize or have such a small difference that it can be taken as reasonably accurate. In an ideal situation, not only the forecast and the corresponding observed value of river stage/ inflow should be the same but also the time of such occurrence should be the same as that predicted.

As per present practice, all the level and inflow forecasts are being judged by the single criteria of accuracy i.e. the actual level attained is within ± 15 cm of forecasted value for stage forecasts and the actual inflow/ volume received in the dam/ barrage is within $\pm 20\%$ of the forecasted value for inflow forecast.

The flood forecasting performance of the level forecasting as well as inflow forecasting sites from 2000 to 2012 is given as Fig1.





List of River Basins(Ref Map –I)

Basin Code	Basin Name
1	Indus (Up to border)
2a	Ganga
2b	Brahmaputra
2c	Barak and others
3	Godavari
4	Krishna
5	Cauvery
6	Subernarekha
7	Brahmani and Baitarni
8	Mahanadi
9	Pennar
10	Mahi
11	Sabarmati
12	Narmada
13	Tapi
14	West flowing rivers from Tapi to Tadri
15	West flowing rivers from Tadri to Kanyakumari
16	East flowing rivers between Mahanadi and Pennar
17	East flowing rivers between Pennar and Kanyakumari
18	West flowing rivers of Kutch and Saurashtra including Luni
19	Area of inland drainage in Rajasthan
20	Minor rivers draining into Bangladesh
21	Minor rivers draining into Myanmar
22	Area of North Ladakh not draining into Indus
23	Drainage Area of Andaman and Nicobar Islands
24	Drainage Area of Lakshadweep Islands

List of Flood Forecasting Stations (Ref Map-I)

Sl. No.	Name of Forecast Station	Sl. No.	Name of Forecast Station	Sl. No.	Name of Forecast Station	Sl. No.	Name of Forecast Station	Sl. No.	Name of Forecast Station
1	Ditrogerh	34	Nani	67	Bella	100	Hayaghat	133	Jalavi Dam
2	Rahakadia	35	Dhansa Regulator	68	Buxar	101	Jharkhandpur	134	Gangakhet
3	Chandmani (Khowang)	36	Delhi Railway Bridge	69	Elgin Bridge	102	Baoua	135	Nanded
4	Nanglamoraghat	37	Mathura	70	Ayodhya	103	Balthara	136	Singur Dam
5	Sbeagar	38	Agra	71	Belampur	104	Kursela	137	Nizamsgar Dam
6	Reemaghat	39	Etawah	72	Bera	105	Sahyogi	138	Srinmagar
7	Bedatghat	40	Gandhinagar Dam	73	Gorakhpur (Bridge)	106	Dangaghat	139	Bhendara
8	Golaghat	41	Auraiya	74	Turtipar	107	Jhawa	140	Pauni
9	Rumalgat	42	Kolpi	75	Darauli	108	Farakka Barrage	141	Baharsha
10	N T Road Crossing (Jaliharail)	43	Hamirpur	76	Gangpur Siwan	109	Massanjore Dam	142	Kateswaram
11	Tespur	44	Mohana	77	Chhapra	110	Tilpara Barrage	143	Jagdlpur
12	Kampur	45	Sahyana	78	Indrapuri	111	Narayanpur	144	Eturunagaram
13	Dheramtil	46	Banda	79	Koelwar	112	Gheropara	145	Dummagudem
14	Gumohati (D C Court)	47	Chilaghat	80	Maner	113	Tenughat Dam	146	Bhadrachalam
15	NH Crossing (Dufiman)	48	Nars	81	Patna (Dighaghat)	114	Ranchet Dam	147	Kunawaram
16	N T Road Crossing (Pagladia)	49	Srinagar	82	Patna (Gandhighat)	115	Mailton Dam	148	Rajahmundry GW Railway Bridge
17	Godopara	50	Rishikesh	83	Khadda	116	Durgapur Barrage	149	Dowlaswaram Barrage
18	Sei Road Bridge	51	Hardwar	84	Chota	117	Hemichole	150	Hemichole Dam
19	NH Crossing (Manes)	52	Narora Barrage	85	Rasipur	118	Kangsabati Dam	151	Naraj
20	Dhubri	53	Moradabad	86	Kamtau	119	Mohampur	152	Alipargal
21	Golokganj	54	Bareilly	87	Sripalpur	120	Dandiwada Dam	153	Nimpara
22	Tufanguri	55	Kannauj (Gumatia)	88	Hathidah	121	Dhans Dam	154	Jenapur Expressway
23	Ghughumari	56	Ainarghat	89	Munger	122	Subah Bridge (Ahmedabad)	155	Anandpur
24	NH 31	57	Kampur	90	Lalbaghighat	123	Kadana Dam	156	Alkhuapada
25	Madhabnagar	58	Delmau	91	Muzzafarpur (Sikandarpur)	124	Wanaboni Weir	157	NH 5 Road Bridge
26	Domohani Road Bridge	59	Phaphamau	92	Revaghat	125	Handia	158	Rajghat
27	Mekhluguri	60	Alahabad (Chitrag)	93	Semastipur	126	Hoshangabad	159	Purushottampur
28	Sonamara	61	Mirzapur	94	Rosara	127	Gandeshwar	160	Gumapur
29	Kalashshar	62	Varanasi	95	Khagaria	128	Bharuch	161	Kashinagar
30	Maduri	63	Hanuman (Sethu/Ludhnow)	96	Bhagalpur	129	Hathur Dam	162	Gotta Barrage
31	Kartipur	64	Jaunpur	97	Colgong/Kahalgaon	130	Ukai Dam	163	Madhuban Dam
32	Annapurmaghat (Sicher)	65	Rae-Bareilly	98	Berabad	131	Surat	164	Daman
33	Tajewala Barrage (Hathodkandi Barrage)	66	Gwalpur	99	Elmghat	132	Kopergaon	165	Vapi Town

List of Flood Forecasting Stations of CWC			
Sl.No.	Name of the river	Name of FF site	Name of State
1	2	3	4
Ganga Basin			
1	Alaknanda	Srinagar	Uttaranchal
2	Ganga	Rishikesh	Uttaranchal
3	Ganga	Haridwar	Uttaranchal
4	Ganga	Narora Barrage	Uttar Pradesh
5	Ganga	Kannauj	Uttar Pradesh
6	Ganga	Ankinghat	Uttar Pradesh
7	Ganga	Kanpur	Uttar Pradesh
8	Ganga	Dalmau	Uttar Pradesh
9	Ganga	Phaphamau	Uttar Pradesh
10	Ganga	Allahabad Chhatnag	Uttar Pradesh
11	Ganga	Mirzapur	Uttar Pradesh
12	Ganga	Varanasi	Uttar Pradesh
13	Ganga	Ghazipur	Uttar Pradesh
14	Ganga	Buxar	Bihar
15	Ganga	Ballia	Uttar Pradesh
16	Ganga	Patna Dighaghat	Bihar
17	Ganga	Patna Gandhighat	Bihar
18	Ganga	Hathidah	Bihar
19	Ganga	Munger	Bihar
20	Ganga	Bhagalpur	Bihar
21	Ganga	Kahalgaon	Bihar
22	Ganga	Sahibgunj	Jharkhand
23	Ganga	Farakka	West Bengal
24	Ramganga	Moradabad	Uttar Pradesh
25	Ramganga	Bareilly	Uttar Pradesh
26	Yamuna	Tajewala Weir	Haryana
27	Yamuna	Mawi	Uttar Pradesh
28	Yamuna	Delhi Rly Bridge	NCT Delhi
29	Yamuna	Mathura	Uttar Pradesh
30	Yamuna	Agra	Uttar Pradesh
31	Yamuna	Etawa	Uttar Pradesh
32	Yamuna	Auraiya	Uttar Pradesh
33	Yamuna	Kalpi	Uttar Pradesh
34	Yamuna	Hamirpur	Uttar Pradesh
35	Yamuna	Chilaghat	Uttar Pradesh
36	Yamuna	Naini	Uttar Pradesh
37	Sahibi	Dhansa	NCT Delhi
38	Chambal	Gandhisagar Dam	Madhya Pradesh
39	Betwa	Mohana	Uttar Pradesh
40	Betwa	Sahjina	Uttar Pradesh
41	Ken	Banda	Uttar Pradesh
42	Gomati	Lucknow HanumanSetu	Uttar Pradesh
43	Gomati	Jaunpur	Uttar Pradesh
44	SAI	Raibareli	Uttar Pradesh
45	Ghaghra	Elqin Bridge	Uttar Pradesh

Sl.No.	Name of the river	Name of FF site	Name of State
1	2	3	4
47	Ghaghra	Turtipar	Uttar Pradesh
48	Ghaghra	Darauli	Bihar
49	Ghaghra	Gangpur Siswan	Bihar
50	Ghaghra	Chhapra	Bihar
51	Rapti	Balrampur	Uttar Pradesh
52	Rapti	Bansi	Uttar Pradesh
53	Rapti	Gorakpur Birdghat	Uttar Pradesh
54	Sone	Inderpuri	Bihar
55	Sone	Koelwar	Bihar
56	Sone	Maner	Bihar
57	PunPun	Sripalpur	Bihar
58	Gandak	Khadda	Uttar Pradesh
59	Gandak	Chatia	Bihar
60	Gandak	Rewaghat	Bihar
61	Gandak	Hazipur	Bihar
62	Burhi Gandak	Lalbeghiaghat	Bihar
63	Burhi Gandak	Muzaffarpur Sikandarpur	Bihar
64	Burhi Gandak	Samastipur	Bihar
65	Burhi Gandak	Rosera	Bihar
66	Burhi Gandak	Khagaria	Bihar
67	Bagmati	Benibad	Bihar
68	Bagmati	Hayaghat	Bihar
69	Adhwara Group	Kamtaul	Bihar
70	Adhwara Group	Ekmighat	Bihar
71	Kamla Balan	Jhanjharpur	Bihar
72	Kosi	Basua	Bihar
73	Kosi	Baltara	Bihar
74	Kosi	Kursela	Bihar
75	Mahananda	Dhengraghat	Bihar
76	Mahananda	Jhawa	Bihar
77	Mayurakshi	Massanjore Dam	Jharkhand
78	Mayurakshi	Tilpara Barrage	West Bengal
79	Mayurakshi	Narayanpur	West Bengal
80	Ajoy	Gheropara	West Bengal
81	Damodar	Tenughat Dam	Jharkhand
82	Damodar	Panchet Dam	Jharkhand
83	Damodar	Durgapur Barrage	West Bengal
84	Barakar	Maithon Dam	Jharkhand
85	Mundeshwari	Harinkhola	West Bengal
86	Kangsabati	Kangsabati Dam	West Bengal
87	Kangsabati	Mohanpur	West Bengal
Brahmaputra Basin			
88	Brahmaputra	Dibrugrah	Assam
89	Brahmaputra	Neamatighat	Assam
90	Brahmaputra	Tezpur	Assam
91	Brahmaputra	Guwahati	Assam
92	Brahmaputra	Goalpara	Assam
93	Brahmaputra	Dhubri	Assam
94	Burhidihing	Naharkatia	Assam

Sl.No.	Name of the river	Name of FF site	Name of State
1	2	3	4
96	Desang	Nanglamoraghat	Assam
97	Dikhow	Shivsagar	Assam
98	Subansiri	Badatighat	Assam
99	Dhansiri (S)	Golaghat	Assam
100	Dhansiri (S)	Numaligarh	Assam
101	Jiabharali	Jiabharali_NTX	Assam
102	Kopilli	Kampur	Assam
103	Kopilli	Dharmatul	Assam
104	Puthimari	Puthimari_NHX	Assam
105	Pagladiya	Pagladia_NTX	Assam
106	Beki	Beki_NHX	Assam
107	Manas	Manas_NHX	Assam
108	Sankosh	Golakganj	Assam
109	Raidak-I	Tufanganj	West Bengal
110	Torsa	Ghughumari	West Bengal
111	Jaldhaka	NH-31	West Bengal
112	Jaldhaka	Mathabhanga	West Bengal
113	Tista	Domohani	West Bengal
114	Tista	Mekhliganj	West Bengal
Barak & Meghna Basins			
115	Barak	APGhat	Assam
116	Katahal	Matizuri	Assam
117	Kushiyara	Karimganj	Assam
118	Manu	Kailashar	Tripura
119	Gumti	Sonamura	Tripura
Eastern Rivers (Excluding Mahanadi)			
120	Subernarekna	Rajghat	Orissa
121	Burhabalang	NH_5_Road Bridge	Orissa
122	Baitarni	Anandpur	Orissa
123	Baitarni	Akhuapada	Orissa
124	Brahmani	Jenapur	Orissa
125	Rushikuluya	Purushottampur	Orissa
126	Vamsadhara	Gunupur	Orissa
127	Vamsadhara	Kashinagar	Orissa
128	Vamsadhara	Gotta Barrage	Andhra Pradesh
Mahanadi Basin			
129	Mahanadi	Hirakud Dam	Orissa
130	Mahanadi	Naraj	Orissa
131	Mahanadi	Alipingal Devi	Orissa
132	Mahanadi	Nimapara	Orissa
Godavari Basin			
133	Godavari	Kopergaon	Maharashtra
134	Godavari	Jaikwadi Dam	Maharashtra
135	Godavari	Gangakhed	Maharashtra
136	Godavari	Nanded	Maharashtra
137	Godavari	Sriram Sagar	Andhra Pradesh
138	Godavari	Kaleswaram	Andhra Pradesh
139	Godavari	Eturunagaram	Andhra Pradesh
140	Godavari	Dummagudam	Andhra Pradesh
141	Godavari	Rhadrachalam	Andhra Pradesh

Sl.No.	Name of the river	Name of FF site	Name of State
1	2	3	4
144	Godavari	Dowalaiswaram	Andhra Pradesh
145	Wardha	Balharsha	Maharashtra
146	Wainganga	Bhandara	Maharashtra
147	Wainganga	Pauni	Maharashtra
148	Manjira	Singur Dam	Andhra Pradesh
149	Manjira	Nizamsagar Dam	Andhra Pradesh
150	Indravati	Jagdulpur	Chhatisgarh
Krishna Basin			
151	Krishna	Arjunward	Maharashtra
152	Krishna	Alamati Dam	Karnataka
153	Krishna	Narayanpur Dam	Karnataka
154	Krishna	Priyadarshini	Andhra Pradesh
155	Krishna	Srisailem Dam	Andhra Pradesh
156	Krishna	Prakasham Barrage	Andhra Pradesh
157	Bhima	Deongaon	Karnataka
158	Tungbhadra	Tungabhadra Dam	Karnataka
159	Tungbhadra	Mantralayam	Andhra Pradesh
Southern River System:			
160	Pennar	Nellore	Andhra Pradesh
Western River Systems:			
161	Banas	Dantiwada Dam	Gujarat
162	Sabarmati	Dharoi Dam	Gujarat
163	Sabarmati	Ahmedabad Shubhash Bridge	Gujarat
164	Mahi	Kadana Dam	Gujarat
165	Mahi	Wanakbori	Gujarat
166	Narmada	Mandla	Madhya Pradesh
167	Narmada	Hoshangabad	Madhya Pradesh
168	Narmada	Garudeshwar	Gujarat
169	Narmada	Bharuch	Gujarat
170	Tapi	Hatnur Dam	Maharashtra
171	Tapi	Ukai Dam	Gujarat
172	Tapi	Surat	Gujarat
173	Damanganga	Madhuban Dam	Gujarat
174	Damanganga	Vapi Town	Gujarat
175	Damanganga	Daman	Dadra & Nagar Haveli